

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10: TRIGONOMETRY I (3.1) FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly? Use what you learned in all ten lessons to answer the prompts below. Write in full sentences, show every step of any calculation, and include units. Draw labelled right-angled triangles wherever they help. Remember the Kisumu data: a student 1.6 m tall casts a 1.2 m shadow, the mango tree casts a 9.0 m shadow, and the sun angle measured with the clinometer is about 53° . Be honest about any rounding or measurement error in your work.

Part 1: Why does the angle tell us about the sides?

In the Kisumu school compound, the student and the mango tree both stand in the same sunlight. Draw and label the two right-angled triangles (one for the student and her shadow, one for the tree and its shadow). Explain why the ratio of opposite to adjacent is the same in both triangles, even though one triangle is much bigger. Then explain why this means that an angle can tell us about the ratios of the sides of a right-angled triangle.

Draw two right-angled triangles. In each one, the vertical side (the student of 1.6 m, or the tree) is the opposite side to the sun angle. The shadow (1.2 m, or 9.0 m) is the adjacent side. The sun's ray is the hypotenuse. Both triangles have a right angle and the same sun angle of about 53° , so their angles are the same and the triangles are similar: same shape, different size. In similar triangles, corresponding sides are in proportion, so $\text{opposite} \div \text{adjacent}$ is the same in both. For the student it is $1.6 \div 1.2 \approx 1.33$. Because of this, the angle alone fixes the ratios $\text{opposite} \div \text{hypotenuse}$, $\text{adjacent} \div \text{hypotenuse}$ and $\text{opposite} \div \text{adjacent}$, whatever the size of the triangle. We named these sine, cosine and tangent. So if we know the angle and one side, we can find another side.

Part 2: Sine, cosine and tangent, and how they are related

(a) State the definitions of $\sin \theta$, $\cos \theta$ and $\tan \theta$ for an acute angle θ in a right-angled triangle. (b) Starting from these definitions, show that $\tan \theta = \sin \theta \div \cos \theta$. (c) Using $\sin 53^\circ \approx 0.7986$ and $\cos 53^\circ \approx 0.6018$

(a) $\sin \theta = \text{opposite} \div \text{hypotenuse}$; $\cos \theta = \text{adjacent} \div \text{hypotenuse}$; $\tan \theta = \text{opposite} \div \text{adjacent}$. (b) $\sin \theta \div \cos \theta = (\text{opposite} \div \text{hypotenuse}) \div (\text{adjacent} \div \text{hypotenuse})$. The hypotenuse cancels, leaving $\text{opposite} \div \text{adjacent}$, which is $\tan \theta$. So $\tan \theta = \sin \theta \div \cos \theta$, and the three ratios are linked. (c) $\sin 53^\circ \div \cos 53^\circ \approx 0.7986 \div 0.6018 \approx 1.327$. This agrees with the table value $\tan 53^\circ \approx 1.3270$, so the relationship holds for the Kisumu sun angle.

from the tables, check this relationship for 53° and say how your answer compares with $\tan 53^\circ \approx 1.3270$.	
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Part 3: Finding the height of the mango tree

(a) Use the sun angle of 53°, the 9.0 m shadow and $\tan 53^\circ \approx 1.3270$ to calculate the height of the mango tree. Show which side is opposite and which is adjacent. (b) Find the height a second way, using the student's height and shadow and similar triangles. (c) Compare your two answers. Why are they slightly different, and what should we say about the tree's height? Explain why you did not need to climb the tree.	(a) The tree height is the side opposite the 53° angle and the shadow is the adjacent side. $\tan 53^\circ = \text{height} \div 9.0$, so $\text{height} = 9.0 \times 1.3270 \approx 11.94$ m. (b) Similar triangles: $\text{tree height} \div 9.0 = 1.6 \div 1.2$, so $\text{height} = 9.0 \times 1.6 \div 1.2 = 12.0$ m. (c) The answers, 11.94 m and 12.0 m, agree to within about 0.1 m. The small difference comes from the measured sun angle being approximate (about 53°) and from rounding. We can say the mango tree is about 12 m tall. Two independent methods agree, which shows the angle-ratio relationship is reliable. We did not climb the tree because the angle and one measured length (the shadow) are enough to find the unknown side using the tangent ratio.
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Part 4: Special angles and complementary angles

(a) Suppose the sun in Kisumu made an angle of exactly 60° with the ground and the mango tree's shadow was still 9.0 m. Using exact values only (no calculator), find the height of the tree. Do the same for an angle of exactly 45°. (b) In the 53° triangle, the other acute angle is 37°. Explain how $\sin 53^\circ$ and $\cos 37^\circ$ are related and why. Write the identity that you used.	(a) For 60°: $\tan 60^\circ = \sqrt{3}$, so $\text{height} = 9.0 \times \sqrt{3} = 9\sqrt{3}$ m, which is about 15.6 m. For 45°: $\tan 45^\circ = 1$, so $\text{height} = 9.0 \times 1 = 9.0$ m. These exact values come from the triangles with sides 1, $\sqrt{3}$, 2 (half of an equilateral triangle of side 2) and 1, 1, $\sqrt{2}$ (an isosceles right-angled triangle). (b) In a right-angled triangle the two acute angles add up to 90°, so they are complementary: $53^\circ + 37^\circ = 90^\circ$. The side that is opposite 53° is adjacent to 37°, and the hypotenuse is shared. So $\sin 53^\circ = \text{opposite} \div \text{hypotenuse} = \cos 37^\circ$. The identity is $\sin \theta = \cos(90^\circ - \theta)$. So $\sin 53^\circ = \cos 37^\circ \approx 0.7986$.
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Part 5: Answering the driving question for any height

(a) Suppose there is no shadow, for example a school	(a) Stand a measured horizontal distance (the baseline) from the base of the flagpole. Use the clinometer to measure the angle of elevation from your eye to the top of the pole. The baseline is the adjacent side and the height is the opposite side, so $\text{height} =$
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flagpole or a Safaricom mast in Kericho. Describe how you could find its height using a clinometer and a tape measure. Name the angle you measure, the length you measure, and the ratio you use. (b) Compare your Lesson 1 hand-drawn ratio estimates with the calculator values. Explain why they might differ, and why it is important to report any difference honestly. (c) Now write a short answer to the driving question and to the two KICD questions: How do we apply trigonometric ratios in day-to-day activities? How are the trigonometric ratios of acute angles related to each other?	baseline $\times$ $\tan(\text{angle of elevation})$. Then add the height of your eye above the ground. This is the same method used for the mango tree, with the baseline replacing the shadow. (b) Hand-drawn estimates differ from calculator values because rulers and protractors have limited precision, so the percentage error reflects measurement limits, not a flaw in the mathematics. Integrity means reporting the real error and not changing the data to match the calculator. (c) A sunbeam, or a line of sight, makes an angle with the ground that forms a right-angled triangle. All triangles with the same angle are similar, so the ratios $\sin$, $\cos$ and $\tan$ depend only on the angle. Knowing the angle and one length lets us calculate any other length, such as the height of a tree, building or hill, without measuring it directly. Day-to-day uses include surveying a plot in Kisumu, finding the height of a mast in Kericho, and an aircraft's glide path at Wilson Airport. The ratios are related: $\tan \theta = \sin \theta \div \cos \theta$, and for complementary angles $\sin \theta = \cos(90^\circ - \theta)$, so they form one connected system.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of similar triangles and ratio ideas	Clearly explains that triangles with the same angle are similar, so the ratios are fixed by the angle alone, and links this to the Kisumu data.	States that the ratios stay the same for the same angle but gives only a partial reason about similarity.	Describes the ratios without explaining why they stay constant, or confuses the sides.
Correct calculations and exact values	Correctly finds the tree height (about 11.94 m and 12.0 m) and the special-angle heights ($9\sqrt{3}$ m and 9.0 m), showing every step with units and exact values.	Calculations are mostly correct with minor slips in steps, units or rounding.	Calculations are incomplete or incorrect, or the opposite and adjacent sides are mixed up.
Relationships among the ratios	Proves $\tan \theta = \sin \theta \div \cos \theta$ and explains $\sin \theta = \cos(90^\circ - \theta)$ using the swap of opposite and adjacent sides.	States both relationships correctly and checks them with numbers, but the explanation is brief.	States a relationship without a reason or with errors.
Application, reasoning and integrity	Describes a complete method for finding any height from an angle and a baseline, explains measurement error honestly, and answers the driving question clearly with Kenyan examples.	Describes a workable method and mentions error, with a general answer to the driving question.	Method is unclear or missing, and error and honesty are not discussed.

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